



U32-C (PS2251-21) Specification

Preliminary Version

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Revision History

Revision	History	Draft Date	Remark
Preliminary	Preliminary version	June 17, 2024	Ben

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1. INTRODUCTION



1.1. Product Overview

Phison PS2251-21 is a USB4 Gen 3x2 NAND Flash Controller. It supports USB4 Gen 3x2, 40Gbps and backward compatible with USB3.2 Gen2, USB 3.2 Gen1, USB 2.0 and USB 1.1. This product supports up to 4 channels/ 16 Flash Chip Enable (CE), sequential read performance is up to 4GB/s and sequential write performance is up to 4GB/s.

1.2. Block Diagram

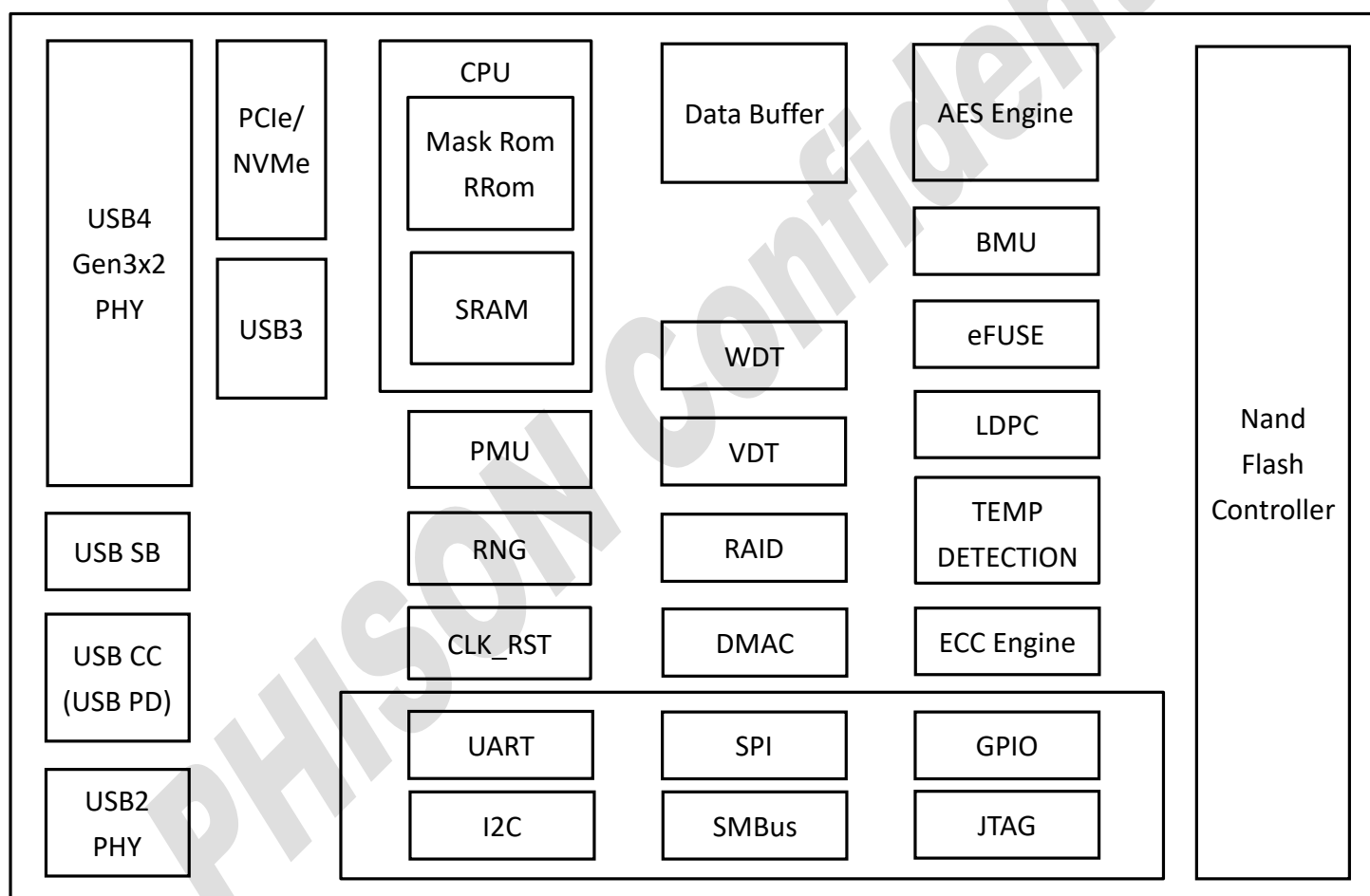


Figure 1-1 System Block Diagram

*PMU: Power Management Unit

1.3. Features Specifications

Features	Specifications
USB	• USB4 Gen 3x2 (40Gbps) • USB 3.2 Gen 2x2 (20Gbps) • Compliance with USB 3.2 Gen1 (5Gbps) • Compliance with USB 2.0 (480Mbps) • Compliance with USB 1.1 (12Mbps)
NAND Flash	• Support up to 4TB • Support Triple-level cell (TLC) and Quad-level cell (QLC) NAND flash memory • Compliant with Toggle 1.0/2.0/3.0/4.0 NAND flash interface • Compliant with ONFI 2.0/3.0/4.1 • Transfer rate: up to 1600 MT/s • Support up to 4 channels/ 16 Chip Enable (CE) within Single Design • Flash IO Operating voltage supply 1.2v/1.8v
Data Reliability	• LDPC ECC protection
OS compatibility	• Support Windows XP and later version • Support MAC OS 10.8 and later version • Support Linux kernel 2.4.10 and later version
Temperature Range	• Operating Range: 0~70 °C • Storage Range: -40~85 °C
Humidity	• 20%~90%
Other Features	• Built-in internal Thermal Sensor

2. PERFORMANCE



Table 1-1 Performance for Kioxia BiCS6

BiCS6		Sequential (MB/s)			Random (MB/s)	
		Read	Write	Sustained	Read	Write
1TB	8CE	4000	3600	700	750	650
2TB	16CE	4000	3600	700	750	650
4TB	16CE	4000	3600	700	750	650

Note:

Performance is measured with the following conditions:

- (a) Sequential : CrystalDiskMark 8.0, 1GB range, QD=Q8T1, Random : CrystalDiskMark 8.0, 1GB range, QD=32T16, Sustained: H2
- (b) Base on USB 4 Gen3x2 (40gbp/s) platform

3. POWER COMSPUTION



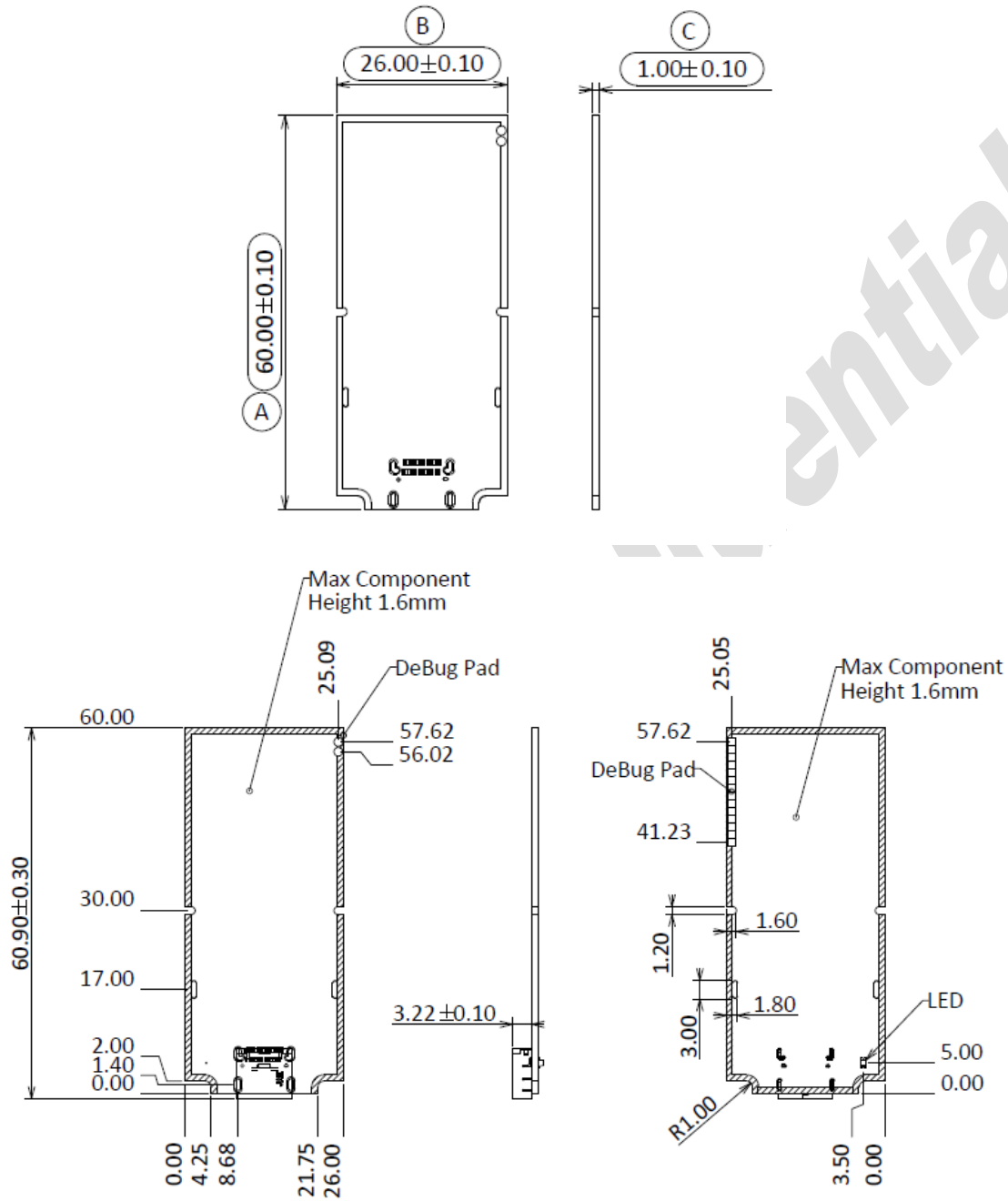
Table 2-1 Power Consumption

Item	Power (mA)
Read	700
Write	780
Idle	252
Suspend	2.4

Note: The above values are for reference only, it may change according to the flash memory used

4. DIMENSION

- Dimension: 60 ± 0.10 mm (L) x 26 ± 0.10 mm (W) x 1 ± 0.10 mm (H)



Notes :

- = Max Component Height
- = No Component
- = No Component / Signal Vias / Signal Copper / Printing
- General Tolerance ± 0.1
- are critical dimensions (A) ~ (C)
- V-Cut area allowed burr : 0.1mm(max.)
- SMT Rounter area allowed tolerance: ± 0.05 mm

NOTES:

- All dimensions are in mm.

5. PRODUCT WARRANTY POLICY



For any other products manufactured and supplied by Phison (“Phison Products”), Phison hereby certify that in the event Phison Product does not conform to the specification for the period ending on the date at which customer’s use of a Phison Product exceeds Phison Product’s total Terabytes Written as recorded by or derived from Phison Product’s S.M.A.R.T. Attribute, including but not limited to Phison Product’s drive life is used up in accordance with the S.M.A.R.T. Attribute, (“Warranty Period”) and such inconformity is confirmed by Phison to be solely attributable to Phison, Phison agrees to repair or replace the nonconforming Phison Product, free of charge.

Notwithstanding the foregoing, the aforementioned warranty shall exclude the inconformity arising from, in relation to or associated with:

- (1) alternation, modification, improper use, misuse or excessive use of Phison Product;
- (2) failure to comply with Phison’s instructions;
- (3) Phison’s compliance with or use of the instructions, technologies, designs, specifications, devices, materials, components, parts, software and firmware provided, instructed or approved by Buyer (including any of its parents, subsidiaries, affiliates, suppliers, subcontractors or downstream customers);
- (4) combination of Phison Product with other materials, components, parts, goods, hardware, firmware or software not supplied by Phison;
- (5) any claim brought by a third party who is commonly known as intellectual property right assertion entity or patent troll;
- (6) NAND flash itself or NAND flash which is embedded into Phison Products;
- (7) Phison’s compliance with general industry standards;
- (8) other error or failure not solely attributable to Phison’s cause (including without limitation, normal wear or tear, manufacturing or assembly wastage, improper operation, virus, unauthorized maintenance or repair).

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6. REFERENCE



The following table is to list out the standards that have been adopted for designing the product.

Table 3-1 List of References

Title	Acronym/Source
USB 4	https://www.usb.org/

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7. TERMINOLOGY



The following table is to list out the acronyms that have been applied throughout the document.

Table 4-1 List of Terminology

Term	Definitions
USB	Universal Serial Bus
SSD	Solid state disk
LDPC	Low Density Parity Check
CE	Chip Enable

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